

Project Synopsis

Project Title: Distributed Adaptive Control for Real-Time Load Balancing in EV Microgrids under Nonlinear Disturbances

1. Introduction & Motivation

Modern distributed energy systems, such as residential electric vehicle (EV) microgrids, frequently encounter severe load imbalances that threaten infrastructure integrity. Traditional fixed-parameter controllers fail to maintain stability under the nonlinear and unpredictable charging profiles of modern battery systems. This project proposes a distributed adaptive control system utilizing nonlinear dynamics and optimization algorithms to achieve resilient, real-time power distribution and load balancing across multiple charging nodes.

2. Problem Statement & Objectives

In a decentralized charging network (e.g., a 15-node EV charging infrastructure sharing a single transformer), the system faces dynamic and unpredictable power demands. The primary challenge is to prevent main transformer saturation and localized overheating.

Objective: To achieve uniform load distribution in real-time under nonlinear disturbances (erratic vehicle arrivals and variable battery draw) by minimizing the load differential across the grid:

$$\Delta L = L_{max} - L_{min}$$

3. Proposed Methodology

The system is modeled as a discrete-time closed-loop feedback control system, operating continuously to evaluate and redistribute power.

System Modeling (State-Space Approach):

The state of each EV charging node is defined as $x_i = [L_i, T_i, P_i]$, representing current load, battery temperature, and power allocated, respectively.

Discrete-time state equation: $x(k+1) = f(x_k, u_k, d_k)$

Nonlinear Disturbance Modeling:

- Real-world unpredictability is simulated using a Logistic Map: $X(n+1) = rXn(1 - Xn)$
- Adaptive Control Strategy:
- The central controller computes error: $e = L_{desired} - L_{actual}$
- Cost function: $J = \sum (Li - L_{avg})^2$
- Adaptive update using SGD: $w(t+1) = wt - \eta \nabla J$
- Actuation (Routing): Dijkstra's algorithm determines optimal power routing paths.

4. Hardware & Software Architecture

Hardware:

- ESP32 microcontrollers as distributed nodes, sensors for feedback, and relays for actuation. Communication via I2C and Wi-Fi/UART.

Software:

- MATLAB/Python for modeling and validation; Proteus for circuit simulation.

5. Expected Outcomes & Stability

The system is expected to converge to a balanced state where ΔL approaches zero, ensuring system stability, reduced oscillations, and protection against overload conditions.